TRANSFORMERS
FINAL EXAMINATION

Electrical Engineering Department

Time: 90 minutes | 20 Questions | 5 points each

INSTRUCTIONS

• Answer all questions by selecting the best answer from the given options.

• Formulas are provided at the end of the exam.

• Show your work for partial credit on calculation problems.

1. A transformer core is driven into saturation during operation. If the primary voltage is increased by 20% beyond the rated value, what is the most probable consequence?

A) The core flux increases by 20%, and magnetizing current remains nearly constant.
B) The core flux increases slightly, but magnetizing current increases dramatically.
C) Both flux and magnetizing current increase proportionally by 20%.
D) The flux decreases due to increased core losses.
E) The magnetizing current decreases as reactance increases.

2. Two identical transformers are connected to the same voltage source. Transformer A supplies a load at 0.85 power factor lagging, while Transformer B supplies a load at 0.85 power factor leading. Both deliver the same real power. Which statement is most accurate regarding their terminal voltages?

F) Terminal voltage of A is lower than B.
G) Terminal voltage of B is lower than A.
H) Both have identical terminal voltages.
I) Cannot be determined without knowing the load current.
J) Terminal voltages depend only on the turns ratio.

3. The primary reason for using laminated silicon steel cores instead of solid steel cores in power transformers is to:

K) Reduce hysteresis losses by decreasing the area of the B-H loop.
L) Minimize eddy current losses by increasing the electrical resistance of flux paths.
M) Increase the magnetic permeability to reduce magnetizing current.
N) Improve mechanical strength to withstand electromagnetic forces.
O) Reduce the cost of core material.

4. If a 60 Hz transformer designed for operation at rated voltage is accidentally connected to a 50 Hz supply of the same voltage magnitude, the most significant consequence will be:

P) Reduced core losses and improved efficiency.
Q) Increased flux density leading to core saturation and excessive magnetizing current.
R) Decreased output voltage proportional to the frequency ratio.
S) No significant change in operation.
T) Improved voltage regulation.

5. During an open-circuit test on a transformer, which parameters are primarily determined?

U) Equivalent resistance and leakage reactance
V) Core loss resistance and magnetizing reactance
W) Copper losses and winding resistance
X) Load losses and efficiency
Y) Short-circuit current and impedance voltage

6. What is the most significant factor that limits the maximum power output of a distribution transformer?

Z) Core saturation
AA) Winding temperature rise due to I^2R losses
BB) Voltage regulation exceeding acceptable limits

CC) Mechanical stress on windings
DD) Insulation breakdown voltage

7. A transformer has a tap changer set at the nominal position. If the tap is moved to the +5% position while the primary voltage remains constant, what happens to the secondary voltage?

EE) Increases by approximately 5%
FF) Decreases by approximately 5%
GG) Remains unchanged
HH) Increases by approximately 10%
II) Cannot be determined without knowing the load

8. Two transformers with identical ratings are operating in parallel. If one transformer has a per-unit impedance of 0.05 + j0.08 pu and the other has 0.06 + j0.10 pu, what is the most probable consequence?

JJ) The transformers will share load equally.
KK) The transformer with lower impedance will carry more load.
LL) Large circulating currents will flow between transformers.
MM) The transformer with higher impedance will overheat.
NN) Both transformers will trip on overcurrent.

9. A 50 kVA, 2400/240 V transformer has the following test results: Open-circuit test (at 240 V): 200 W, 2.5 A. Short-circuit test (at 120 V): 600 W, 20.8 A. What is the approximate efficiency at 75% of full load with 0.80 power factor lagging?

OO) 96.2%
PP) 97.1%
QQ) 97.8%
RR) 98.4%
SS) 98.9%

10. An autotransformer with common winding of 200 turns and series winding of 50 turns is used as a step-up transformer. Compared to a conventional two-winding transformer with the same turns ratio, the autotransformer's power rating advantage is approximately:

TT) 1.25
UU) 4.0
VV) 5.0
WW) 8.0
XX) Cannot be determined

11. If a transformer's core has a small air gap intentionally introduced, compared to an identical transformer without an air gap, which statement is most accurate?

YY) Transformer with air gap has higher magnetizing current and lower stored energy.
ZZ) Transformer with air gap has higher magnetizing current and higher stored energy.
AAA) Transformer with air gap has lower magnetizing current and higher flux density.
BBB) Both transformers have identical magnetizing currents and flux densities.
CCC) Transformer with air gap has reduced hysteresis losses.

12. A 100 kVA, 11000/400 V transformer has copper losses of 1.5 kW at full load and core losses of 0.8 kW. What is the approximate load at which the transformer operates at maximum efficiency?

DDD) 53 kVA
EEE) 65 kVA
FFF) 73 kVA
GGG) 87 kVA
HHH) 100 kVA

13. In a Y-Δ connected three-phase transformer bank, if the line voltage on the Y side is 13.8 kV and the turns ratio (Y-side phase winding to Δ-side phase winding) is 20:1, what is the approximate line voltage on the Δ side?

III) 346 V
JJJ) 400 V
KKK) 460 V
LLL) 690 V
MMM) 800 V

14. A transformer is operating at 80% of rated voltage and 60% of rated load. Neglecting losses, if the load power factor is 0.90 lagging, the approximate voltage regulation compared to operation at rated voltage and full load (also 0.90 lagging) will be:

NNN) Significantly higher
OOO) Slightly higher
PPP) Approximately the same
QQQ) Slightly lower
RRR) Significantly lower

15. The primary disadvantage of using an autotransformer instead of a conventional two-winding transformer is:

SSS) Lower efficiency
TTT) Larger physical size
UUU) Lack of electrical isolation between primary and secondary
VVV) Higher core losses
WWW) Cannot operate with different voltage ratios

16. A 25 kVA, 2400/240 V transformer has the following parameters referred to the high-voltage side: Req = 3.0 Ω, Xeq = 5.0 Ω. The transformer delivers rated kVA at 240 V and 0.80 power factor lagging. The approximate voltage regulation is:

XXX) 2.8%
YYY) 3.4%
ZZZ) 4.1%
AAAA) 4.7%
BBBB) 5.3%

17. A distribution transformer is observed to be overheating during normal operation within its rated capacity. Given that all cooling systems are functioning properly, what is the most probable cause?

CCCC) Excessive harmonic content in the load current
DDDD) Low power factor of the load
EEEE) Elevated ambient temperature
FFFF) Incorrect tap setting
GGGG) Low supply voltage

18. For a transformer operating in per-unit system, if the base power is 100 kVA, base voltage (primary) is 2400 V, and the transformer has Zeq = 0.04 + j0.06 pu, the actual equivalent impedance referred to the primary side is approximately:

HHHH) 2.30 + j3.46 Ω
IIII) 0.96 + j1.44 Ω
JJJJ) 4.80 + j7.20 Ω
KKKK) 9.60 + j14.4 Ω
LLLL) 19.2 + j28.8 Ω

19. A Δ-Δ connected three-phase transformer bank loses one of its three transformers. The remaining two transformers are reconnected in an open-delta (V-V) configuration. The capacity of this configuration compared to the original three-transformer bank is approximately:

MMMM) 33%
NNNN) 50%
OOOO) 58%
PPPP) 67%
QQQQ) 87%

20. A transformer experiences a sudden short circuit at its secondary terminals. The most immediate effect on the primary current will be:

RRRR) Limited to rated current due to core saturation
SSSS) Increases dramatically, limited primarily by leakage impedance
TTTT) Remains unchanged due to voltage regulation
UUUU) Decreases due to back emf
VVVV) Oscillates at twice the supply frequency

FORMULAS PROVIDED

Transformer Voltage and Current:

$V_p/V_s = N_p/N_s = a$

$I_p/I_s = N_s/N_p = 1/a$

Impedance Transformation:

$Z' = a^2Z$

Voltage Regulation:

$VR = [(V_{no\text{-}load} - V_{full\text{-}load}) / V_{full\text{-}load}] \times 100\%$

Efficiency:

$\eta = P_{out} / (P_{out} + P_{losses}) \times 100\%$

Per-Unit System:

$Z_{base} = V^2_{base} / S_{base}$

$Z_{pu} = Z_{actual} / Z_{base}$

Three-Phase:

$V_L = \sqrt{3} \times V_\Phi$ (for Y connection)

$V_L = V_\Phi$ (for Δ connection)

$S_{3\Phi} = \sqrt{3} \times V_L \times I_L$

Autotransformer:

Power advantage = (NC + NSE) / NSE

Maximum Efficiency:

Occurs when: $P_{core} = P_{copper}$

Load fraction for max efficiency = $\sqrt{(P_{core} / P_{copper,FL})}$

ANSWER KEY

1. B
2. A
3. B
4. B
5. B
6. B
7. B
8. B
9. B (97.1%)
10. C (5.0)
11. B
12. C (73 kVA)
13. B (400 V)
14. E
15. C
16. C (4.1%)
17. A
18. A (2.30 + j3.46 Ω)
19. C (58%)
20. B